Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/14/2021
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Individual Differences in the Neuronal and Behavioral Response to Exposure to HZE Particles		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rabin@umbc.edu	**Fax:**	FY (410) 455-1055
PI Organization Type:	UNIVERSITY	**Phone:**	(410) 952-1761
Organization Name:	University of Maryland, Baltimore County		
PI Address 1:	Department of Psychology		
PI Address 2:	1000 Hilltop Cir		
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	08/01/2016	**End Date:**	07/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2022 per L. Stephenson/Space Radiation (Ed., 8/2/21) NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 5/4/2020)		
Key Personnel Changes/Previous PI:	May 2020 report: CoInvestigator (CoI) Shibu Poulouse, Ph.D., is no longer CoI on the project.		
COI Name (Institution):	Shukitt-Hale, Barbara Ph.D. (US Department of Agriculture/Agricultural Research Service, Beltsville, MD)		
Grant/Contract No.:	NNX16AE06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Future missions in space (such as a mission to Mars or the establishment of a moon base) will involve long-term travel beyond the magnetic field of the Earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles, and particles of high energy and charge (HZE particles). Previous research has shown that exposure to single, above threshold doses of HZE particles and protons can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements or may produce degenerative changes in neuronal and behavioral functioning after the conclusion of the mission. The effects of exposure to multiple, subthreshold doses of the same or different particles or protons have not been extensively studied. The objectives of the proposed research program are: (1) to evaluate the effects of repeated below threshold/above threshold exposures to the same and different HZE particles and protons on cognitive and neuronal deficits and how they relate to the characteristics of the specific particle; (2) to determine the relationship between changes in performance and HZE particle- and proton-induced changes in oxidative stress, inflammation, signaling molecules, and autophagy in key brain regions; and (3) experiments currently underway will examine the capacity of dietary antioxidants (blueberries) administered after exposure to 56Fe particles to ameliorate the disruptive effects of irradiation on cognitive performance and changes in neurochemical functioning. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and protons and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Although the typical experimental design involves exposing rats to an above threshold dose of different HZE particles, the most likely pattern of radiation exposure to which astronauts will be exposed on exploratory class missions involves repeated exposures to below threshold doses. In this context, threshold is defined as the lowest dose of a specific particle which will produce a deficit in cognitive performance. The current experiments using subthreshold threshold doses of 4He particles in combination with 56Fe particles and reversible combinations of 4He and 1H suggest that the effects on cognitive performance may be task, particle, and sex specific. In operant performance combining 4He particles with exposure to 56Fe particles following a 72 hr delay interval resulted in a significant increase in responsiveness to changes in reinforcement contingencies in female subjects compared to non-irradiated controls and a significant decrease in responsiveness to changes in reinforcement contingencies in male subjects. In contrast there were no consistent effects on performance of the novel object recognition or novel spatial location tasks. With regard to combined exposure to 1H and 4He particles separated by a 2-hr delay interval, the effects were somewhat more consistent. Preexposure to subthreshold doses of 4He particles prevented the disruption of performance on both novel object recognition and novel spatial location tasks following exposure to 50 cGy of protons but increased the probability of a disruption of cognitive performance following exposure to 25 cGy. The mechanisms underlying these opposing effects of preexposure to 4He particles on measures of learning are not certain. In contrast, preexposure to 1H did not consistently modify the effects of exposure to 4He particles on either task. With regard to performance on the operant task, preexposure to either particle caused an increase in responsiveness to changes in reinforcement contingencies from the lower dose (0.015 cGy 4He or 25 cGy 1H).
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Articles in Other Journals or Periodicals	Rabin BM, Cahoon DS, Shukitt-Hale B. "Response to comments by Bevelacqua and Mortazavi." Life Sci Space Res. 2020 Nov;27:111-2. https://doi.org/10.1016/j.lssr.2020.09.003 , Nov-2020
Articles in Peer-reviewed Journals	Shuryak I, Brenner DJ, Blattnig SR, Shukitt-Hale B, Rabin BM. "Modeling space radiation induced cognitive dysfunction using targeted and non-targeted effects." Sci Rep. 2021 Apr 23;11(1):8845. https://doi.org/10.1038/s41598-021-88486-z ; PMID: 33893378; PMCID: PMC8065206 , Apr-2021
Articles in Peer-reviewed Journals	Cahoon DS, Shukitt-Hale B, Bielinski DF, Hawkins EM, Cacioppo AM, Rabin BM. "Effects of partial- or whole-body exposures to 56Fe particles on brain function and cognitive performance in rats." Life Sci Space Res. 2020 Nov;27:56-63. https://doi.org/10.1016/j.lssr.2020.07.006 , Nov-2020